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Classifying Lower-Grade Glioma and Glioblastoma Multiforme Using XGBoost

Introduction. A glioma is a tumor that can form in either the brain or the spinal cord. A glioma can be either benign or malignant. Glioma arises from the uncontrolled growth of glial cells within the central nervous system. Symptoms of glioma include visual disturbances or acute vision loss, difficulty walking or impaired balance, and dizziness [1]. Glioma is a major health concern in the United States, as from 1975 – 2018, 62,159 patients were diagnosed with glioma, and from 1995 – 2018, 31,922 deaths occurred from glioma [2].

Lower-Grade Gliomas are grade I or grade II gliomas that usually affects children and young adults. Although Lower-Grade Gliomas are typically less harmful than higher-grade gliomas, Lower-Grade Gliomas remain difficult to treat, and patient survival outcomes are limited. Furthermore, grade II gliomas frequently transform into higher-grade gliomas [3]. Glioblastoma Multiforme is a grade IV glioma that is the most prevalent type of malignant glioma among adults. Glioblastoma Multiforme can occur age any age; however, Glioblastoma Multiforme most frequently develops in patients that are age 45 to age 70 [4]. Glioblastoma Multiforme is one of the most difficult to treat brain tumors, and Glioblastoma Multiforme often exhibits resistance to conventional treatments such as chemotherapy and radiation therapy [5]. The five-year survival rate of Glioblastoma Multiforme is around 5% [4]. From 1975 – 2018, there were 32,986 cases of Glioblastoma Multiforme, and from 1995 – 2018, there were 21,201 deaths from Glioblastoma Multiforme [2].

Grading patients' glioma tissue is essential for guiding treatment decisions to combat glioma, as it helps doctors decide what treatment procedure a patient should undergo. Treatment procedures for glioma patients include undergoing surgery to remove the glioma tissue then undergoing chemotherapy and radiation therapy to remove remaining cancerous cells, and the "wait-and-see approach" (where doctors monitor a patient's glioma without providing immediate active treatments to address the glioma, and only recommend active treatments when the state of the patient's glioma warrants them). The wait-and-see approach is more commonly employed for patients with Lower-Grade Gliomas. The "gold standard" for glioma grading is microscopic examination of a patient's glioma tissue that is extracted through biopsy. However, this process can be time-consuming, invasive, and subject to sampling error. [6]. Machine learning poses a viable alternative to traditional glioma tissue grading, as machine learning is non-invasive, has the potential to produce more accurate results, and can support more personalized glioma treatment strategies [7]. As a result, this study attempts to develop a machine learning model to classify glioma tissues among a cohort of 839 patients part of The Cancer Genome Atlas Program as being Lower-Grade Glioma or Glioblastoma Multiforme [8]. The research question this study attempts to answer is:

Among a cohort of patients part of The Cancer Genome Atlas Program with glioma, how well can machine learning classify the patients as having either Lower-Grade Glioma or Glioblastoma Multiforme?

Data. This study attempts to develop a machine learning model classifying patients as having either Lower-Grade Glioma or Glioblastoma Multiforme using a dataset of 839 patients with glioma from the UC Irvine Machine Learning Repository [8] that was uploaded in 2022. The dataset was compiled from The Cancer Genome Atlas Program, a landmark cancer genomics initiative that molecularly characterized over 20,000 primary cancer and matched normal samples across 33 cancer types [9]. The dataset contains

23 features corresponding to three clinical features and the top 20 most frequently mutated genes in glioma tissues. Features include gender, age at diagnosis, race, and whether the patient has a mutated isocitrate dehydrogenase 1 gene or not (mutations in isocitrate dehydrogenase are frequently detected in patients with Lower-Grade Glioma and Glioblastoma Multiforme [10]). The outcome variable is glioma grade, which corresponds to whether a patient has Lower-Grade Glioma (the negative class) or Glioblastoma Multiforme (the positive class). The dataset contained no missing values. Below is a table summarizing the characteristics of the cohort of patients:

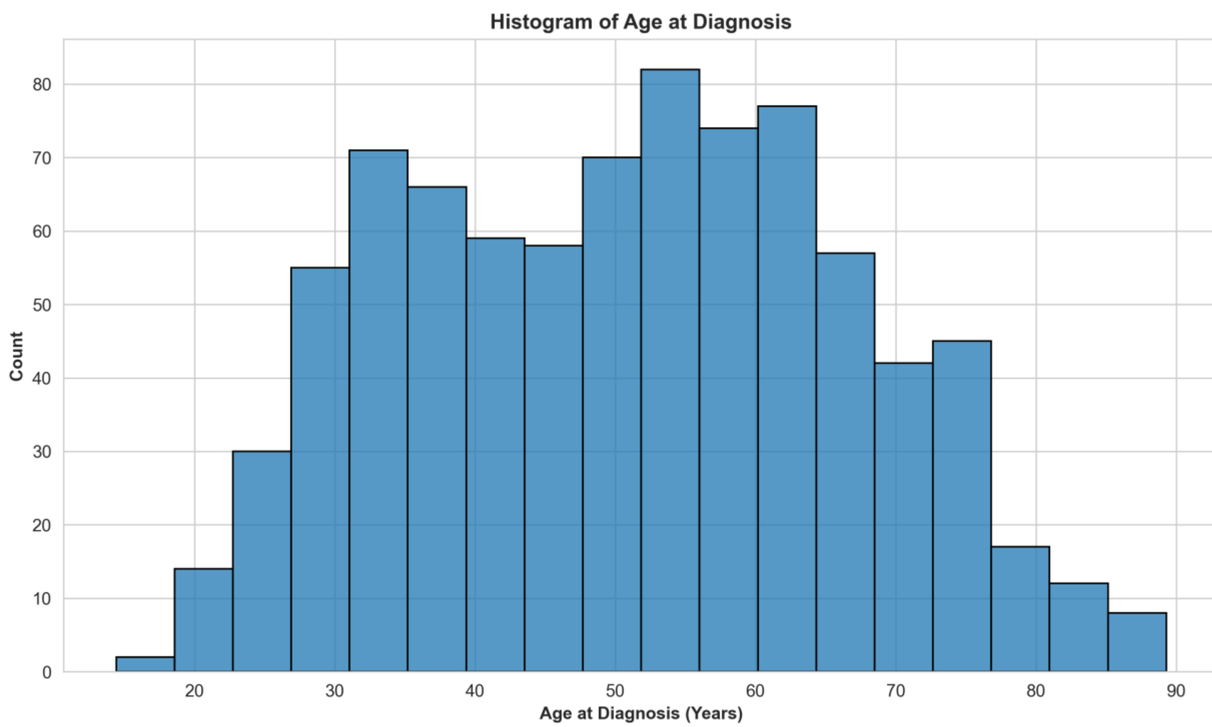
Table 1. Summary characteristics of 839 patients with glioma that took part in The Cancer Genome Atlas Program, stratified by glioma grade

	Lower-Grade Glioma	Glioblastoma Multiforme
Overall, count (%)	487 (58.05)	352 (41.95)
Gender, count (%)		
Male	271 (55.65)	217 (61.65)
Female	216 (44.35)	135 (38.35)
Race, count (%)		
White	457 (93.84)	308 (87.50)
Black or African American	21 (4.31)	38 (10.80)
Asian	8 (1.64)	6 (1.70)
American Indian or Alaska Native	1 (0.21)	0 (0)
Isocitrate Dehydrogenase 1, count (%)		
Not Mutated	106 (21.77)	329 (93.47)
Mutated	381 (78.23)	23 (6.53)
Tumor Protein P53, count (%)		
Not Mutated	252 (51.75)	239 (67.90)
Mutated	235 (48.25)	113 (32.10)
ATRX Chromatin Remodeler, count (%)		
Not Mutated	304 (62.42)	318 (90.34)
Mutated	183 (37.58)	34 (9.66)
Phosphatase and Tensin Homolog, count (%)		
Not Mutated	462 (94.87)	236 (67.05)
Mutated	25 (5.13)	116 (32.95)
Epidermal Growth Factor Receptor, count (%)		
Not Mutated	456 (93.63)	271 (76.99)
Mutated	31 (6.37)	81 (23.01)
Capicua Transcriptional Repressor, count (%)		
Not Mutated	380 (78.03)	348 (98.86)
Mutated	107 (21.97)	4 (1.14)
Mucin 16, Cell Surface Associated, count (%)		
Not Mutated	446 (91.58)	295 (83.81)
Mutated	41 (8.42)	57 (16.19)

Phosphatidylinositol-4,5-Bisphosphate 3-Kinase Catalytic Subunit Alpha, count (%)		
Not Mutated	448 (91.99)	318 (90.34)
Mutated	39 (8.01)	34 (9.66)
Neurofibromin 1, count (%)		
Not Mutated	458 (94.05)	314 (89.20)
Mutated	29 (5.95)	38 (10.80)
Phosphoinositide-3-Kinase Regulatory Subunit 1, count (%)		
Not Mutated	466 (95.69)	319 (90.63)
Mutated	21 (4.31)	33 (9.38)
Far Upstream Element Binding Protein 1, count (%)		
Not Mutated	444 (91.17)	350 (99.43)
Mutated	43 (8.83)	2 (0.57)
RB Transcriptional Corepressor 1, count (%)		
Not Mutated	481 (98.77)	318 (90.34)
Mutated	6 (1.23)	34 (9.66)
Notch Receptor 1, count (%)		
Not Mutated	449 (92.20)	352 (100)
Mutated	38 (7.80)	0 (0)
BCL6 Corepressor, count (%)		
Not Mutated	470 (96.51)	340 (96.59)
Mutated	17 (3.49)	12 (3.41)
CUB and Sushi Multiple Domains 3, count (%)		
Not Mutated	475 (97.54)	337 (95.74)
Mutated	12 (2.46)	15 (4.26)
SWI/SNF Related, Matrix Associated, Actin Dependent Regulator of Chromatin, Subfamily A, Member 4, count (%)		
Not Mutated	464 (95.28)	348 (98.86)
Mutated	23 (4.72)	4 (1.14)
Glutamate Ionotropic Receptor NMDA Type Subunit 2A, count (%)		
Not Mutated	480 (98.56)	332 (94.32)
Mutated	7 (1.44)	20 (5.68)
Isocitrate Dehydrogenase 2, count (%)		
Not Mutated	466 (95.69)	350 (99.43)
Mutated	21 (4.31)	2 (0.57)

FAT Atypical Cadherin 4, count (%)		
Not Mutated	476 (97.74)	340 (96.59)
Mutated	11 (2.26)	12 (3.41)
Platelet-Derived Growth Factor Receptor Alpha, count (%)		
Not Mutated	481 (98.77)	336 (95.45)
Mutated	6 (1.23)	16 (4.55)
Age at diagnosis, mean (SD)	43.87 (13.25)	60.70 (13.41)

Figure 2. Histogram of distribution of age of diagnosis for 839 patients with glioma that took part in The Cancer Genome Atlas Program



The above histogram conveys that the dataset covers a fairly wide variety of values for patient age at diagnosis. The age at diagnosis for the glioma patients in the dataset ranges from around 14.5 years to around 89.5 years. The above distribution of patient age at diagnosis contains around 2 peaks. One peak is at approximately 33 years, and other peak is at approximately 54 years.

Methods. For this study, the machine learning algorithm XGBoost was applied to the dataset to classify patients as either having Low-Grade Glioma or Glioma Multiforme. XGBoost was developed as an optimized extension of Gradient Boosted Trees. Gradient Boosted Trees is an ensembled learning method that builds decision trees sequentially using the technique of boosting. Gradient Boosted Trees starts off by fitting an initial "weak learner" decision tree to the data that often performs only slightly better than random guessing. At each training iteration of the algorithm, Gradient Boosted Trees computes the negative gradient of the loss function with respect to current predictions to help create a decision tree that attempts to correct the errors made by the existing ensemble. XGBoost is also an ensemble learning method that builds decision trees sequentially using boosting. Several modifications were made to the Gradient Boosted Trees algorithm in order to attempt to make XGBoost computationally faster, have stronger predictive performance, more scalable to larger datasets, and more resistant to overfitting than Gradient Boosted Trees [11]. Below is the regularized objective function that XGBoost attempts to minimize at each training iteration:

$$(1)\ \mathcal{L}^{(t)} = \sum_{i=1}^{n} \ell\left(y_i,\ \hat{y}_i^{(t-1)} + f_t(x_i)\right) + \Omega(f_t)$$

where *n* = the number of rows in the dataset, *t* = the current iteration number, x_i = the feature vector for row *i* in the dataset, f_t = the decision tree that was trained at iteration *t*, y_i = the true label for row *i* in the dataset, and $\hat{y}_i^{(t-1)}$ = the existing ensemble's prediction for row *i* that is calculated before adding f_t to the ensemble. $\sum_{i=1}^{n} \ell\left(y_i,\ \hat{y}_i^{(t-1)} + f_t(x_i)\right)$ corresponds to the loss function that is used by both Gradient Boosted Trees and XGBoost. The loss function measures how much the ensemble's predictions differ from the true labels. $\Omega(f_t)$ represents the additional regularization term that is used by XGBoost only [11].

The purpose of XGBoost's regularization term $\Omega(f_t)$ is to penalize each decision tree trained at each iteration for being too complex so that the overall ensemble of decision trees can generalize better to unseen data. The regularization term helps to prevent each decision tree from memorizing noise or variations specific only to the training data, and instead incentivizes each decision tree to learn overall patterns in the training data that are relevant to future datasets. Below is XGBoost's regularization term presented more in-depth [11]:

$$(2)\ \Omega(f_t) = \gamma T + \frac{1}{2}\lambda \sum_{j=1}^{T} w_j^2 + \alpha \sum_{j=1}^{T} |w_j|$$

where *T* = the number of leaves in a decision tree trained at iteration *t*; w_j = the weight of leaf j of a decision tree trained at iteration *t*; and γ, λ, α each are tunable hyperparameters representing the "cost" assigned to each term in the regularization expression. Each term in the regularization expression penalizes a different aspect of the complexity of each trained decision tree. The γT term discourages each decision tree from having too many leaves. The $\frac{1}{2}\lambda \sum_{j=1}^{T} w_j^2$ term discourages each leaf of each decision tree from having a large weight w_j. The $\alpha \sum_{j=1}^{T} |w_j|$ term also discourages each leaf of each decision tree from having a large weight w_j; however, this term is different in that it can shrink weights to being exactly 0. It is important to note that $\alpha \sum_{j=1}^{T} |w_j|$ is an optional penalty term that isn't included in all implementations of XGBoost [12].

Second-order Taylor approximation is typically used to solve XGBoost's regularized objective function (1) in a more computationally feasible manner. Below is the simplified regularized objective function:

$$(3)\ \mathcal{L}^{(t)} \approx \sum_{i=1}^{n} \left[\ell\left(y_i, \hat{y}^{(t-1)}\right) + g_i f_t(x_i) + \frac{1}{2} h_i f_t(x_i)^2 \right] + \Omega(f_t)$$

where $g_i = \partial_{\hat{y}^{(t-1)}} \ell\left(y_i, \hat{y}^{(t-1)}\right)$ is the first order gradient of the loss function, and $h_i = \partial^2_{\hat{y}^{(t-1)}} \ell\left(y_i, \hat{y}^{(t-1)}\right)$ is the second order gradient (Hessian) of the loss function [11].

One reason why XGBoost is an appropriate algorithm to use for this study's classification task of classifying patients as either having Lower-Grade Glioma or Glioblastoma Multiforme is XGBoost contains built-in support to address the issue of overfitting through the regularization term in XGBoost's regularized objective function. Given the context of trying to classify patients' glioma, it is important to avoid overfitting as best as possible, as an algorithm that performs poorly on unseen patient data could systematically miss many high-risk patients, potentially leading to fatal consequences. Furthermore, an algorithm that performs poorly on unseen patient data could end up causing many patients to be recommended the incorrect treatment plan for their glioma through their glioma grade being misclassified. Given that the dataset only contains 839 patients and a smaller cohort size increases the risk of overfitting, XGBoost is particularly useful because it helps to mitigate this increased risk of overfitting. XGBoost is also a strong algorithm to use for this study's task of classifying patients' gliomas because XGBoost is adept at capturing complex, non-linear interactions in data [13]. Since the clinical and gene features in this study's glioma dataset are likely to have complex, non-linear interactions, XGBoost is an effective algorithm to employ for this study's glioma classification task. In addition, XGBoost is a suitable algorithm because XGBoost has been demonstrated to perform well in classification tasks for healthcare [14].

To prepare the glioma dataset before applying XGBoost model on it, the "age at diagnosis" feature was standardized to follow the standard normal distribution and the "race" feature was one hot encoded. In addition, the glioma dataset was split so that 70% was allotted to the training set and 30% was allotted to the test set. Hyperparameter tuning was applied to the training data in order to find the optimal set of parameters to use for the XGBoost model. The optimal set of parameters was decided by selecting the set of parameters that produced the highest 5-fold cross-validated AUC on the training dataset. Afterward, the XGBoost model with the optimal parameters was applied on the test set. The optimally tuned XGBoost model's performance was evaluated on the training set using mean 5-fold cross validated error metrics of AUC, accuracy, precision, and recall. In addition, the optimally tuned XGBoost model's performance was measured on the test set using error metrics of AUC, accuracy, precision, and recall.

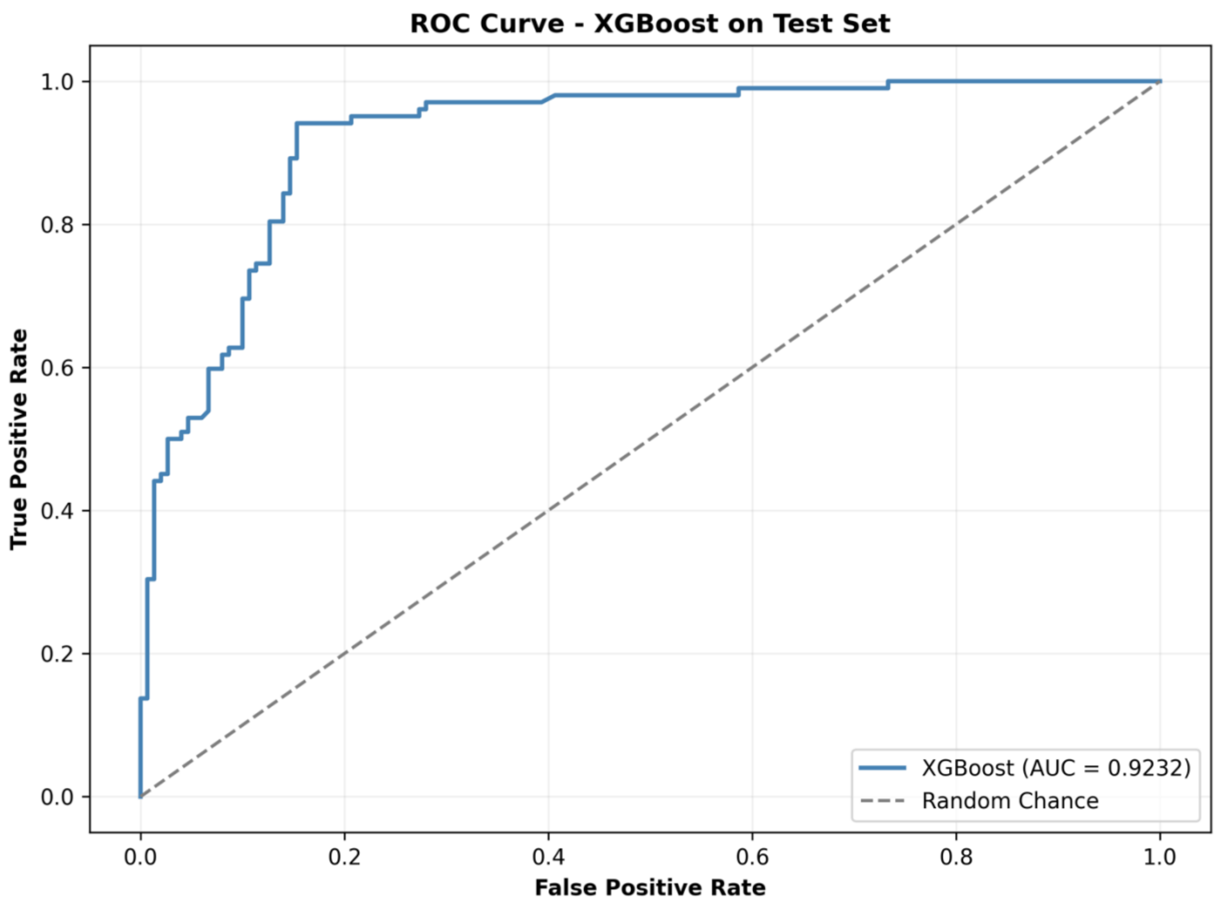
Results. Below is a table comparing the performance of the optimally tuned XGBoost model on the training set and on the test set:

Table 2. Error metrics of optimally tuned XGBoost model applied on training set and on test set

	Training Set	Test Set
AUC	0.9179	0.9232
Accuracy	0.8586	0.8849
Precision	0.7943	0.8067
Recall	0.9005	0.9412

Below is the ROC curve for the optimally tuned XGBoost model applied on the test set:

Figure 2. ROC curve for the optimally tuned XGBoost model applied on the test set



Discussion. Overall, XGBoost performed fairly well for the task of classifying patients as either having Lower-Grade Glioma or Glioblastoma Multiforme. XGBoost was able to achieve a very strong AUC of 0.9232 on the test set. This means that XGBoost was able to distinguish between the two different classes very well, which is advantageous given the classification context of attempting to distinguish Lower-Grade Glioma from Glioblastoma Multiforme. XGBoost also achieved a fairly high accuracy of 0.8849 on the test set, meaning that XGBoost was able to make lots of correct predictions for glioma grade. This is a clinically meaningful result, as misclassifying a patient's glioma grade could lead to an inappropriate treatment recommendation. XGBoost also achieved a high recall score on the test set of 0.9412, demonstrating that XGBoost was able to identify a fairly high amount of cases of Glioblastoma Multiforme out of all of the existing cases of Glioblastoma Multiforme. This is really valuable in a clinical context given how difficult it is to cure Glioblastoma Multiforme, as missing Glioblastoma Multiforme cases could result in fatal consequences. XGBoost was also able to achieve a fairly high precision score of 0.8067 on the test set, which means that out of the Glioblastoma Multiforme classifications that XGBoost made, XGBoost was able to detect a fairly large amount of true Glioblastoma Multiforme cases. However, 0.8067 still means that a fairly significant amount of patients with Lower-Grade Glioma (approximately 19.33%) were incorrectly classified to have Glioblastoma

Multiforme. In a clinical context, this is meaningful because Lower-Grade Glioma patients incorrectly classified as Glioblastoma Multiforme could be directed toward unnecessarily aggressive treatment, including intensive chemotherapy and radiation, which carry significant side effects and reduced quality of life. A future step is to improve the XGBoost model's precision score further in order to minimize the amount of patients with Lower-Grade Glioma that are incorrectly recommended to undergo intensive treatment. It is important to note that across all four error metrics, the XGBoost model's performance was higher on the test set than the training set. This means that the trained XGBoost model generalized well to unseen data. This is a particularly encouraging result in a clinical context, as a model that generalizes well is more likely to produce reliable classifications when applied to new patient cohorts beyond the current patient cohort in The Cancer Genome Atlas Program dataset.

Overall, XGBoost shows promise as a machine learning tool to classify glioma patients as either having Lower-Grade Glioma or Glioblastoma Multiforme that can serve as a viable alternative to traditional biopsy methods. XGBoost was able to achieve fairly high AUC, accuracy, and recall on the test set. Furthermore, XGBoost was able to generalize well to the test set. A crucial component of XGBoost's performance is XGBoost was able to achieve a high recall score, which is critical because it means that XGBoost was able to detect lots of patients to have the more fatal disease of Glioblastoma Multiforme. However, more work needs to be done to improve the precision score of XGBoost in order to prevent XGBoost from recommending the incorrect treatment plan by misclassifying patients with Lower-Grade Glioma to have Glioblastoma Multiforme.

Works Cited

[1] “Glioma.” Cleveland Clinic, Cleveland Clinic, 3 Sept. 2025, my.clevelandclinic.org/health/diseases/21969-glioma.

[2] Lin, D., Wang, M., Chen, Y., Gong, J., Chen, L., Shi, X., Lan, F., Chen, Z., Xiong, T., Sun, H., & Wan, S. (2021). Trends in Intracranial Glioma Incidence and Mortality in the United States, 1975-2018. Frontiers in oncology, 11, 748061. https://doi.org/10.3389/fonc.2021.748061

[3] Youssef, G., & Miller, J. J. (2020). Lower Grade Gliomas. Current neurology and neuroscience reports, 20(7), 21. https://doi.org/10.1007/s11910-020-01040-8

[4] “Glioblastoma (GBM).” Cleveland Clinic, Cleveland Clinic, 23 Apr. 2025, my.clevelandclinic.org/health/diseases/17032-glioblastoma.

[5] Yalamarty, S. S. K., Filipczak, N., Li, X., Subhan, M. A., Parveen, F., Ataide, J. A., Rajmalani, B. A., & Torchilin, V. P. (2023). Mechanisms of Resistance and Current Treatment Options for Glioblastoma Multiforme (GBM). Cancers, 15(7), 2116. https://doi.org/10.3390/cancers15072116

[6] Pereira, S., Meier, R., Alves, V., Reyes, M., Silva, C.A. (2018). Automatic Brain Tumor Grading from MRI Data Using Convolutional Neural Networks and Quality Assessment. In: Stoyanov, D., et al. Understanding and Interpreting Machine Learning in Medical Image Computing Applications. MLCN DLF IMIMIC 2018 2018 2018. Lecture Notes in Computer Science(), vol 11038. Springer, Cham. https://doi.org/10.1007/978-3-030-02628-8_12

[7] Liu S, Xie Y, Gong X, He J, Zou W (2025) Machine learning-based prediction of glioma grading. PLoS One 20(12): e0314831. https://doi.org/10.1371/journal.pone.0314831

[8] Tasci, E., Zhuge, Y., Kaur, H., Camphausen, K., & Krauze, A. V. (2022). Hierarchical Voting-Based Feature Selection and Ensemble Learning Model Scheme for Glioma Grading with Clinical and Molecular Characteristics. International Journal of Molecular Sciences, 23(22), 14155.

[9] “The Cancer Genome Atlas Program (TCGA).” National Cancer Institute Center for Cancer Genomics, U.S. Department of Health and Human Services, www.cancer.gov/ccg/research/genome-sequencing/tcga. Accessed 15 June 2026.

[10] Liu, Y., Lang, F., Chou, F. J., Zaghloul, K. A., & Yang, C. (2020). Isocitrate Dehydrogenase Mutations in Glioma: Genetics, Biochemistry, and Clinical Indications. Biomedicines, 8(9), 294. https://doi.org/10.3390/biomedicines8090294

[11] Tianqi Chen and Carlos Guestrin. 2016. XGBoost: A Scalable Tree Boosting System. In Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD '16). Association for Computing Machinery, New York, NY, USA, 785–794. https://doi.org/10.1145/2939672.2939785

[12] Guillaume, S. (2022, February 24). Visually understand XGBoost, LightGBM and CatBoost Regularization Parameters. Towards Data Science. https://towardsdatascience.com/visually-understand-xgboost-lightgbm-and-catboost-regularization-parameters-aa12abcd4c17/

[13] https://medium.com/@athi.9307/why-xgboost-works-so-well-for-nonlinear-patterns-trend-changes-and-missing-data-8358c279978e

[14] Rafie, Z., Talab, M. S., Koor, B. E. Z., Garavand, A., Salehnasab, C., & Ghaderzadeh, M. (2025). Leveraging XGBoost and explainable AI for accurate prediction of type 2 diabetes. BMC public health, 25(1), 3688. https://doi.org/10.1186/s12889-025-24953-w